SYSTEM ARCHITECT ARM CORTEX A

Bài 3.3: Sensor Driver (Regmap & IIO)

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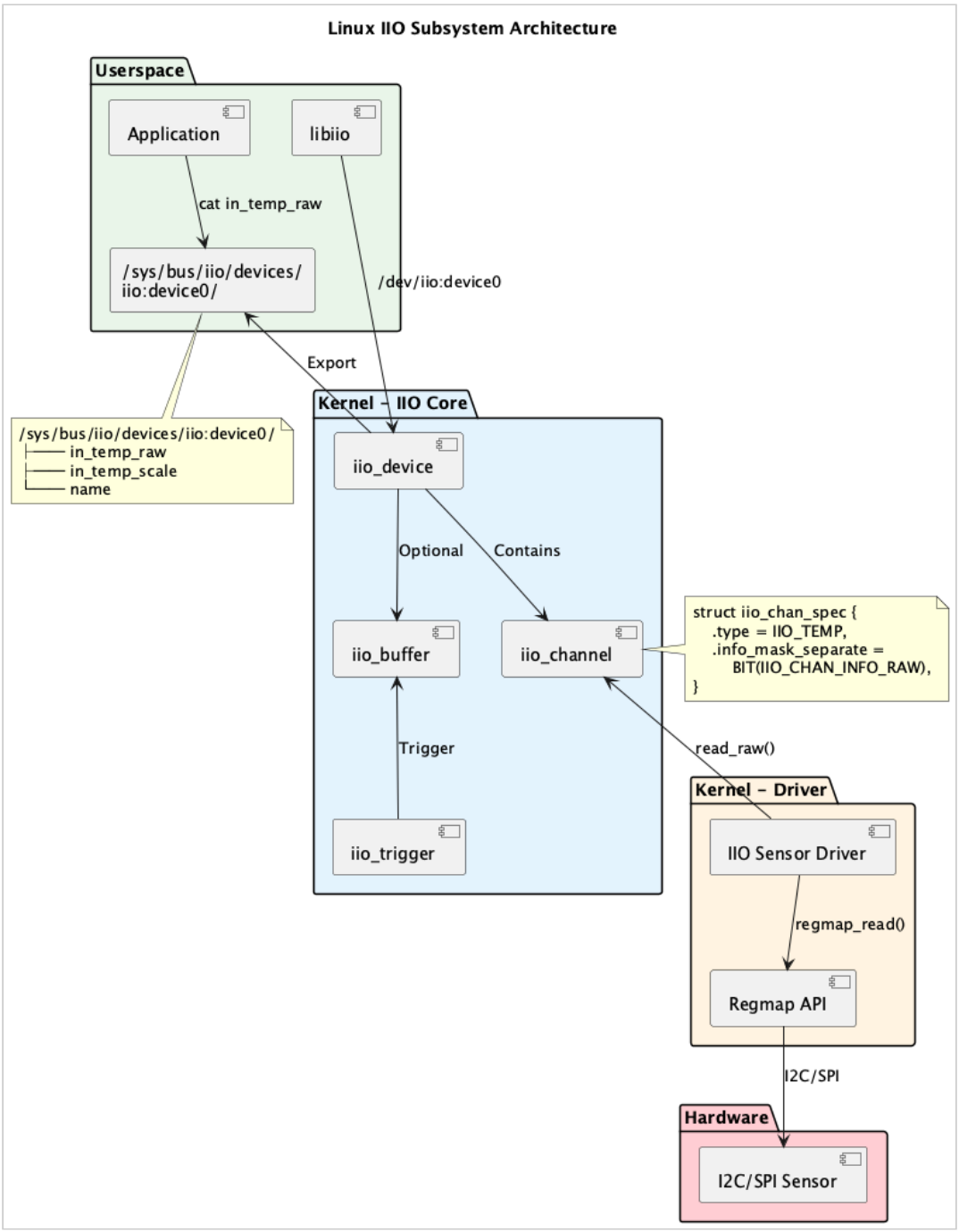
Mục tiêu Bài học

Sau buổi học này, học viên sẽ có khả năng:

- Hiểu **IIO (Industrial I/O) Subsystem**
- Sử dụng **Regmap API** cho I2C/SPI registers
- Viết **IIO sensor driver**

Phần 1: IIO Subsystem



Hình 1: Kiến trúc IIO Subsystem

1.1 IIO là gì?

- **IIO:** Subsystem cho sensors (ADC, DAC, accelerometer, gyro, light...)
- Standard sysfs interface

- Buffer support (continuous sampling)
- Trigger mechanism

1.2 IIO Channel Types

Type	Macro	Mô tả
Voltage	IIO_VOLTAGE	ADC input
Current	IIO_CURRENT	Current sensor
Accel	IIO_ACCEL	Accelerometer
Gyro	IIO_ANGL_VEL	Gyroscope
Light	IIO_LIGHT	Ambient light
Temp	IIO_TEMP	Temperature

Phần 2: Regmap API

2.1 Tại sao dùng Regmap?

- Abstract hóa I2C/SPI register access
- Caching, locking built-in
- Debug via debugfs

2.2 Regmap Config

```
static const struct regmap_config my_sensor_regmap_config = {
    .reg_bits = 8,
    .val_bits = 8,
    .max_register = 0xFF,
    .cache_type = REGCACHE_RBTREE,
};

// In probe
```

```
struct regmap *regmap;
regmap = devm_regmap_init_i2c(client, &my_sensor_regmap_config);
```

2.3 Regmap Operations

```
// Read single register
int val;
regmap_read(regmap, REG_STATUS, &val);

// Write single register
regmap_write(regmap, REG_CTRL, 0x01);

// Read-modify-write
regmap_update_bits(regmap, REG_CTRL, MASK_ENABLE, MASK_ENABLE);

// Bulk read
u8 buf[6];
regmap_bulk_read(regmap, REG_DATA, buf, 6);
```

Phần 3: IIO Driver Structure

3.1 IIO Channel Definition

```
static const struct iio_chan_spec my_sensor_channels[] = {
    {
        .type = IIO_ACCEL,
        .modified = 1,
        .channel2 = IIO_MOD_X,
        .info_mask_separate = BIT(IIO_CHAN_INFO_RAW),
        .info_mask_shared_by_type = BIT(IIO_CHAN_INFO_SCALE),
        .scan_index = 0,
        .scan_type = {
            .sign = 's',
            .realbits = 16,
            .storagebits = 16,
            .endianness = IIO_LE,
        },
    },
    // Y and Z channels similar...
};
```

3.2 Read Function

```
static int my_sensor_read_raw(struct iio_dev *indio_dev,
                              struct iio_chan_spec const *chan,
                              int *val, int *val2, long mask)
{
    struct my_sensor_data *data = iio_priv(indio_dev);
    __le16 raw;

    switch (mask) {
    case IIO_CHAN_INFO_RAW:
        regmap_bulk_read(data->regmap,
                         REG_ACCEL_X + chan->scan_index * 2,
                         &raw, sizeof(raw));
        *val = sign_extend32(le16_to_cpu(raw), 15);
        return IIO_VAL_INT;

    case IIO_CHAN_INFO_SCALE:
        *val = 0;
        *val2 = 9810;  // 0.009810 m/s^2 per LSB
        return IIO_VAL_INT_PLUS_MICRO;
```

```
    default:
        return -EINVAL;
    }
}
```

3.3 Probe Function

```
static int my_sensor_probe(struct i2c_client *client)
{
    struct iio_dev *indio_dev;
    struct my_sensor_data *data;

    indio_dev = devm_iio_device_alloc(&client->dev, sizeof(*data));
    data = iio_priv(indio_dev);

    data->regmap = devm_regmap_init_i2c(client, &my_sensor_regmap_config);

    indio_dev->name = "my-sensor";
    indio_dev->modes = INDIO_DIRECT_MODE;
    indio_dev->channels = my_sensor_channels;
    indio_dev->num_channels = ARRAY_SIZE(my_sensor_channels);
    indio_dev->info = &my_sensor_info;

    return devm_iio_device_register(&client->dev, indio_dev);
}
```

Phần 4: Sysfs Interface

```
# List IIO devices
ls /sys/bus/iio/devices/

# Read raw values
cat /sys/bus/iio/devices/iio:device0/in_accel_x_raw
cat /sys/bus/iio/devices/iio:device0/in_accel_y_raw
cat /sys/bus/iio/devices/iio:device0/in_accel_z_raw

# Read scale
cat /sys/bus/iio/devices/iio:device0/in_accel_scale

# Calculate real value: raw * scale
```

Phần 5: Device Tree

```
&i2c1 {
    status = "okay";

    accelerometer@1d {
        compatible = "vendor,my-sensor";
        reg = <0x1d>;
        interrupt-parent = <&pio>;
        interrupts = <7 5 IRQ_TYPE_EDGE_RISING>;  // PH5
    };
};
```

Phần 6: Câu hỏi Ôn tập

1. IIO subsystem dùng cho loại device nào?
2. Regmap có những ưu điểm gì?
3. IIO_CHAN_INFO_RAW và IIO_CHAN_INFO_SCALE khác nhau?
4. Làm sao đọc sensor qua sysfs?
5. scan_type trong iio_chan_spec có ý nghĩa gì?

Tài liệu Tham khảo

- IIO Documentation
- Regmap API
- IIO Tools (libiio)

Yêu cầu Bài tập

- Hiểu IIO channel structure

- Sử dụng Regmap cho I2C access
- Đọc sensor qua sysfs interface